

APM Project Management Qualification

Learner Study Pack



Life cycles

Introduction

This section of the Learner Study Pack will cover the Life cycles competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of life cycles as a framework comprising a set of distinct high-level stages required to transform an idea of concept into reality in an orderly and efficient manner.

Learning objective and outcomes

Learning objective:

1. Understand the distinct life cycle stages used to structure and organise a project

By the end of this activity, you should be able to:

- 1a) Understand the distinctive features of linear, iterative and hybrid life cycles (including why projects are structured as phases in linear life cycles) and know when each is applicable.
- 1b) Knowledge of the differences between a project life cycle and an extended life cycle.
- 1c) Understand how the context and culture of an organisation, and the needs of a specific project, influence the choice of project life cycle and any adaptations that may be needed to the life cycle.
- 1d) Knowledge of the strengths and limitations of different life cycles.

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Life cycles:

- APM Body of Knowledge Seventh edition 1.2 (Life cycle options and choices)
- APM (2019) Hybrid project management: is it your next challenge? <https://www.apm.org.uk/blog/hybrid-project-management-is-it-your-next-challenge/>
- APM Which life cycle approach is right for you? <https://www.apm.org.uk/resources/what-is-project-management/what-is-a-life-cycle/>

Overview

Life cycles offer a systematic and organised way to undertake project-based work and can be viewed as the structure underpinning deployment.

There are several life cycle approaches that an organisation can adopt to deliver beneficial change through a project. The approach chosen will determine the way projects, programmes and portfolios operate and play a vital role in determining their success.

Approaches range from highly predictive models such as a linear approach, assuming well established knowledge in regards the context through to highly adaptive models such as iterative which operate in more uncertain, volatile and dynamic environments.

Features of Different Life cycles

Outlined below are the main life cycles and their distinctive features:

Life cycle	Features	Benefits	Strengths	Limitations
Linear	Sequential with clearly defined outputs. Applies to stable low-risk projects with greater structure, where expectations of each phase are known.	Provides a clear framework for the team to follow, with early definition of requirements and allows maximum control and governance. Controls and information are passed on to the next phase when predefined milestones have been reached and accomplished.	Suitable for low-risk projects. Provides a clear framework for the team to follow.	Requires very early clarity on scope and governance. Longer process to ascertaining benefits realised. Less flexible to accommodating change.
Iterative	Repeats one or more phases. Project scope might be vague, or solutions unclear requiring greater flexibility in governance procedures.	Beneficial for evolving objectives or solutions, used in agile development projects and allows iterative learning. Project manager and team can observe the benefits the staged functionality delivers and adjust the next iteration accordingly.	Beneficial for evolving objectives or solutions, and agile projects. Suitable where project scope might be vague. Allows iterative feedback and accommodates change more easily.	Lack of early certainty in terms of overall duration and cost. Could lead to complexities in managing resource.

Life cycle	Features	Benefits	Strengths	Limitations
Hybrid	Adds iteration to a linear life cycle enabling a mix of approaches.	Project team can choose the best of both linear and iterative life cycles to best suit the project. If a fixed requirement is set, the iterative style can be used during deployment to develop the output and add further functionality.	Facilitates a mix of approaches to suit a specific development. Building agile working into a project or programme can offer increased efficiency and flexibility.	It requires great skill and clarity when using multiple different systems of working to be successful.

Phases and Linear Life cycles

Projects are structured as phases in linear life cycles because it:

- Provides clearer identification of priorities and appropriate focus on the work which is currently taking place.
- Achieves early understanding of requirements through spending adequate time developing the initial idea or a detailed plan.
- Delivers more effective stakeholder communication, and regular project reviews facilitating frequent consultation and transparency.
- Achieves maximum transparency by assessment of achievements against requirements at end of phase (predefined milestones).
- Allows for maximum control and governance over the project.

Differences between a Project Life cycle and an Extended Life cycle

Project Life cycle	Extended Life cycle
A project life cycle contains the phases up to transition and close. A business delivering the project on behalf of a customer may use the standard project life cycle, as they are likely to complete the contract at the handover of the project to the customer (end user). Accountability for the output is handed over.	An extended life cycle goes beyond the transition and close phase to include the adoption of outputs and the benefits realisation phase. This is suitable where the project is expected to incorporate management of change and benefit realisation. The extended life cycle is suited to projects that are funded within the organisation and are making a change to the business. The project team supports the embedding of the project. The team can determine the immediate benefits realised as the project goes operational and through immediate feedback to the organisation on the realisation of benefits. Governance and accountability for adoption of the output stays within the project until the change is fully embedded, thus preventing knowledge boundaries between project teams and operations.

Influencing Factors on Chosen Life cycle

The context and culture of an organisation, as well as the needs of a specific project can have a huge influence on the choice of project life cycle and any adaptations needed.

Organisations may favour linear life cycles when there is:

- A highly structured, predictable and stable process is needed which offers transparency and maximum control and governance.
- Availability of relatively perfect knowledge upfront.
- Resistance to change and inflexibility in terms of corrections and rework.
- Knowledge / teams that are divided into distinct phases, creating silos and knowledge barriers between the phases, particularly when different delivery agents will deliver different phases.
- Lower appetite for risk.

Organisations may favour iterative life cycles when there are:

- Development projects that are agile.
- Departments that are flexible, adaptable, and open to change.
- Concurrency, or simultaneous engineering, where different development steps are allowed to be performed in parallel.
- Uncertainty regarding the scope by allowing the objectives to evolve throughout the life cycle as learning and discovery take place.
- Prototypes, timeboxes or parallel activities utilised to acquire new insights, obtain feedback or explore high-risk options.

Scenario One:

The scale and complexity of the 'Snappy Appy' renovation project is such that choosing the appropriate life cycle will be one of the key factors in determining the project's ultimate success.

'Cultural fit' is always at the heart of the decisions that the organisation makes and Ayo, the Founder/CEO, wants the senior leadership to exemplify this approach as well.

They have asked you to recommend a life cycle that not only ensures that the renovation is a success, but also considers the context and culture of 'Snappy Appy' as well.

Which life cycle would you recommend? Outline the ways in which it aligns with the project context and culture of the company.

Scenario Two

Ayo wants to ensure that the entire senior leadership team are onboard from the start with all the processes and procedures that will be used to ensure that the renovation project goes smoothly.

They have passed your recommended life cycle and reasoning onto the rest of the team for their information.

However, Robin the Finance Manager and Sandeep the Head of Development are both very detailed focused and would like more information about the life cycles that you discounted and your reasoning.

Write a brief email to them both outlining the other life cycles options, along with the reasons why you do not think that they are appropriate for the renovation project.

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the life cycle section:

- In what context would an extended life cycle be a suitable approach to a project and why?
- Why is a phase structure crucial in linear life cycles?

Exam practice questions

1.1 Which combination of stages are all in the project life cycle?
(1 mark)

1. Adoption of outputs
2. Benefits realisation
3. Definition
4. Deployment
5. Concept
6. Post-project review

Select the correct combination of three statements:

- a. 3,5,6
- b. 1,4,6
- c. 2,5,6
- d. 3,4,5

1.2 You are the Project Manager on a long and complex vehicle manufacturing project. The full set of requirements are still being developed.

In this instance, which lifecycle would not be suitable?
(1 mark)

- a. Extended
- b. Linear
- c. Iterative
- d. Hybrid

1.3 Which statement describes the strengths of a linear project lifecycle?
(1 mark)

- a. This lifecycle has the expectation that each phase is known and provides a clear framework for the team to follow.
- b. This lifecycle repeats one or more of the phases of a project or programme before proceeding to the next one.
- c. This lifecycle allows the project manager and team to develop functions of the project as it progresses.
- d. This lifecycle is beneficial for a project where there are evolving objectives or solutions.

1.4 You are a project manager working to deliver a regulatory change. The project has a high risk of legal penalties and needs to be very tightly controlled.

The regulator has provided you with a clear expectation of the changes required and wants to see regular updates on the project's progress and how you are managing the risk to the customer.

Which project life cycle would you use to manage this project and why?

Your response should:

- Clearly state the life cycle you would use to manage this project (1 mark)
- State three reasons for selecting your chosen life cycle (3 marks)
- State one reason why you would not use an alternative life cycle to manage this project (1 mark)

1.5 What are **three** strengths and **two** limitations of choosing an iterative life cycle for a project?
(5 marks)

1.6 What are the two additional phases added to a project which has an extended life cycle?
(2 marks)

1.7 Select the correct option from the drop-down list.

You are the project manager of a large IT development project. There are high levels of uncertainty concerning the final deliverables.

___(a)___ life cycle would be most appropriate for this project. The use of this life cycle will help to ___(b)___.

(2 marks)

(a)

1. A linear
2. A hybrid
3. An iterative
4. An evolutionary

(b)

1. Clearly identify priorities
2. Avoid changes to deliverables
3. Strengthen the Business Case
4. Decide if the project is viable

1.8 When choosing the right life cycle for a project, it is important to consider the deployment approach, the desired benefits and outcomes of the project, and the organisation's appetite for ___(a)___.

More predictive life cycles rely on the team having a lot of knowledge at the start of the project, while ___(b)___ life cycles facilitate knowledge being created as the work progresses.

(2 marks)

(a)

1. Risk
2. Conflict
3. Underspend
4. Change

(b)

1. Waterfall
2. Iterative
3. Linear
4. Extended

Exam practice answers

Question	Answer / Mark Scheme
1.1	D
1.2	B
1.3	A
1.4	1 mark should be awarded for correctly stating a linear life cycle. 1 mark should be awarded for each reason for selecting the chosen life cycle up to a maximum of 3 marks, for example, but not limited to: • Providing greater control. • Providing greater management of risk. • Providing more opportunities to review and make decisions. • Providing more predictability. • Providing a greater opportunity to deliver the correct scope. • Providing a higher probability of high quality. • The fact that requirements and outcomes are known upfront. 1 mark should be awarded for any reason not to use an iterative life cycle, for example, but not limited to: • Too much risk. • The requirements/outcomes are already clearly defined. • There's no need to build prototypes or iterate a new solution. • Regular reporting to the regulator is required.
1.5	1 mark for any strength, 3 marks available in total – for example, but not limited to, the below: • Accommodates a vague project scope. • Allows project objectives to evolve and change over time. • Allows solution to evolve and change over time. • More flexible / more able to integrate change. • Early feedback can be gained and used to improve the overall solution. • Parts of the solution can be delivered early, to enable some benefits earlier. 1 mark for any limitation, 2 marks available in total – for example, but not limited to, the below: • Project duration can be uncertain. • Project costs / budget can be uncertain. • Managing resources can be more complex. • Business case can be difficult to justify if scope is too vague.
1.6	1 mark for each of the below, 2 marks in total: • Adoption/Adoption of outputs • Benefits Realisation
1.7	(a) 3 (b) 1
1.8	(a) 1 (b) 2